

## ps-3 spread Locator.

(1) What is Statistical Distribution?

→ A statistical distribution shows how the values of a variable are spread or occur in a dataset. It describes the frequency or probability of ~~different~~ different values.

(2) What is Q-Q plot and why it used?

→ A Q-Q (Quantile-Quantile) plot compares the distributions of sample data with a theoretical distribution, usually normal. It is ~~mainly~~ used to check whether data <sup>mainly</sup> is approximately normally distributed.

(3) Difference Between Discrete and continuous distributions.

→ Discrete: Takes countable values such as 0, 1, 2, 3 Example: number of students

continuous: - can take any values in a range. Example: height, weight or temperatures.

(4) What is Bernoulli distribution?

→ Bernoulli distribution describes a single experiment with only two possible outcomes: success (1) or Failure (0). Example: a coin toss.

(5) What is Binomial Distribution?

→ Binomial Distribution gives the probability of getting a specific number of successes fixed number of independent Bernoulli trials. Example: number of heads in 10 coin tosses.

(6) Explain Log-Normal Distribution.

→ A log-normal distribution is a ~~distribution~~ ~~Dist~~ ~~a~~ Distribution in which the logarithm of a variable is normally distributed. It is commonly used for too positive, right-skewed data such as income and some biological measurements.

(7) Explain power <sup>Law</sup> ~~Law~~ Distribution?

→ A power law distribution shows that a small number of

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values were very large while many values were small. It followed a relationship such as  $PC \propto x_1$ .  
Example: website popularity, city sizes.

(8) What is Box-Cox Transform?

→ Box-Cox transformation is a method used to transform data to make it more normally distributed and stabilize variance. It is useful before applying statistical tests that assume normality.

(9) Explain poisson Distribution with an example.

→ poisson Distribution gives the probability of a certain number of events occurring in a fixed interval when events occur independent at a known average rate. Example: number of calls received by a call center in one hour.

(10) What is z-score probability?

→ A <sup>Score</sup> ~~z-score~~ tells how many standard deviations a value is from the mean. It is calculated as  $z = \frac{(x - \mu)}{\sigma}$ .

The z-score can be used with the standard Normal Distribution to find probability.

(11) Differentiate probability Density Function (PDF) and Cumulative Distribution Function (CDF).

→ PDF: Shows the relative likelihood or Density of values at different points. For continuous data, probability over an interval is the area under the PDF.

CDF: Gives the probability that a random is less than or equal to a particular value:  $F(x) = P(X \leq x)$ .